Name: Aditya Pathak
Roll no: 2510IA004,
Inft (A)

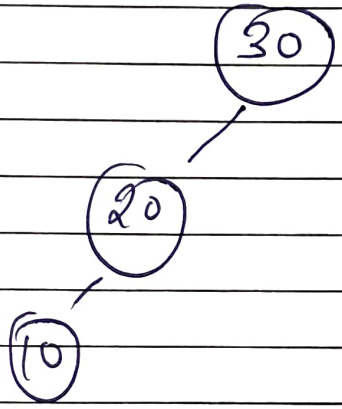
Assignment 04

Q1]

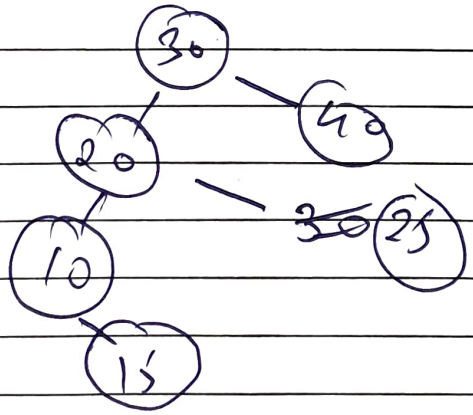
→

Pre : 30, 20, 10, 15, 25, 40, 35, 50

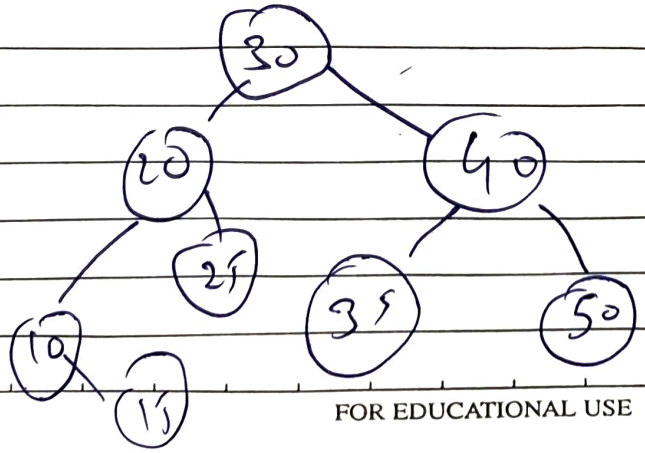
step 1:



step 2:



step 3:



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Postorder traversal: 15, 10, 25, 20, 35, 50, 40

Q2]

→ In a strictly binary tree, every internal node has exactly 2 children.

Total no. of nodes = n
no. of leaf nodes = l

Total no. of nodes = n+l
Total no. of edges = (n+l) − 1

∵ Each internal node has 2 children
No. of edges = 2n

∴ 2n = (n+l) − 1
∴ 2n = n + l − 1
∴ l = 2n − n − 1
l = n + 1

(n+1) leaf nodes

Q3] →

```
#include <stdio.h>
#include <stdlib.h>
typedef struct node
{ int data;
  struct node *left, *right;
} node;

typedef struct
{ node *root;
} BST;
```

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```
void inset (BST *t, int ch)
{ node *p, *q
  p=(node *) malloc (sizeof (node));
  p->data = ele;
   p-> left = p-> right = NULL;
if (t->root == NULL)
{   t-> root = p;
    return;
  }

int countnods (node *t)
{
  if (t == NULL)
    return 0;
  return 1 + countnods (t-> left) +
   countnods (t-> right);
  }

int countleaf (node *t)
{ if (t== NULL)
      return 0;
   if (t-> left == NULL && t-> right == NU
      return 1.
  return countleaf (t-> leaf) + countleaf
  - t->rgh == NULL)
  }

int countBranch (node *t) {
      if ( t == NULL)
          return 0;
   if (t-> left == NULL && t-> right == NULL
      return 0;
```

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```
return 0;
return 1 + countbranch(t->left) + countbranch(t->right);
}

int findDepth(node *t)
{
  int ld, rd;
  if (t==NULL)
    return 0;
  ld = findDepth(t->left);
  rd = findDepth(t->right);

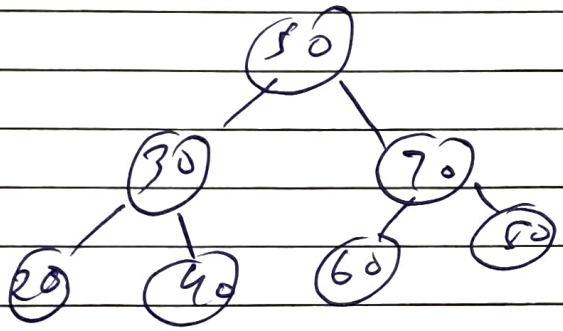
 if (ld > rd)
   return ld+1;
 else
   return rd+1;
 }
void decendg(node *t)
 {
   if (t==NULL)
    return;
  decendg(t->right);
  printf("%d", t->data);
   decendg(t->left);
   }
```

Sample Output :-

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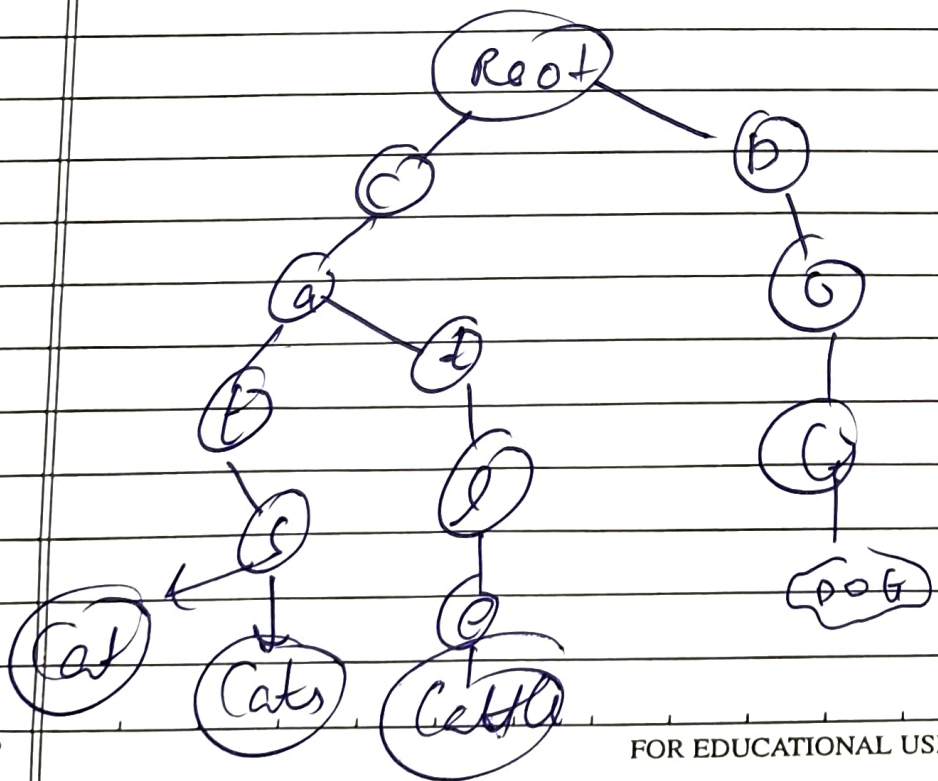
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Roll no: 2510104004
Inst (A)

Enter no. of Node : 7
Enter element : 50 30 20 40 70 60 80
Total node : 7
Leaf node : 4
Branch node : 3
Depth of tree : 3
Level order : 80 70 60 50 40
30 20



Q-4]

→ The data structure used for this is Trie also known as Prefix tree



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Why this → not used?

→ ① BST is designed for ordered search, not prefix search. It may require traversing multiple nodes and many don't share common prefix

- Trie, is the solution as it stores character along the edges and allows efficient prefix search.

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Student Undertaking for Ethical Academic Practice

Vidyalankar Institute of Technology, Mumbai

Assignment No: 07

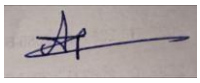
Branch: INFT Vertical: A
Semester: Third(3) Division: A
Subject Name: Data Structure Subject Code: PCIT1T

I hereby declare that for the assignment / academic activity submitted by me, I confirm ONE of the following statements by ticking (☐) the appropriate box:

☐✓ (1) I have prepared the assignment / academic activity entirely by myself.
☐ (2) I have completed the assignment / academic activity with academic guidance or discussion from seniors / friends / peers, while ensuring the work is my own.
☐ (3) I have used AI-based tools only as an aid and have appropriately acknowledged or cited their use as per the ethical guidelines of the Institute.
☐ (4) I have used AI-based tools and directly copy-pasted content for the assignment / academic activity.
☐ (5) I have copied the assignment / academic activity from peers / friends.

I understand that selecting options (4) or (5) indicates unethical academic practice and may attract academic penalties, including rejection of the assignment or disciplinary action, as per the rules of Vidyalankar Institute of Technology, Mumbai. I submit this declaration truthfully and accept full responsibility for the same.

Student Name: Aditya Pathak Roll No.: 25101A0041

Signature:  Date: 03/10/2026